

Realization and Control of a Prototype of Legged Rover for Planetary Exploration

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Abstract—This paper concerns the development of a prototype of a six-legged robot for space exploration. The robot is a testbed for a new control technique based on a peculiar kind of artificial neural network. Accounts are given on the shape and structure of the hexapod, and a brief description of its electronic circuit and computer interface is shown. The paper ends with the results of the tests on two different control systems (a standard one and the neural network based) and the comparison between the properties of the two methods.

I. INTRODUCTION

In this work the development of a prototype of legged rover for planetary exploration [1] will be shown, together with the first results of the control of its motion. Until now the robots aimed at planetary exploration were based on wheels, as that configuration allows a better stability. However a design based on wheels is limited in performances over rough terrain. On the other hand a design based on legs will allow to face more difficult areas allowing to explore steep terrain as well.

The prototype has been developed with the intention to test an innovative algorithm for the control of legged robot based on Continuous Time Recurrent Neural Network (CTRNN) [2]. The use of such kind of network will allow a series of improvements over classical techniques used for the control of a legged robot [3]. The overall design of the prototype has been carried on after a previous numerical work in which it has been proven that the CTRNN networks are extremely robust versus uncertainties and noise and they allow to easily manage the whole mechanical complexity. Moreover, such kind of controller does not require an high level of precision on state measurements.

The developed prototype mimics a six-legged arthropod in which each leg has three rotational joints acted by DC motors; this makes the robot a system with 18 degrees of freedom. The mechanical and structural features of the robot are discussed in section II.

The control of the robot requires to measure 18 different angles. The DC motors are controlled with PWM techniques,

while the sensors used are angular potentiometers mounted on the axis of rotation of each joint, which provide analog measures; this leads to the necessity to design an ad hoc circuit for interfacing actuators and sensors to the computer where the control software will be executed. Microcontrollers are used both to generate PWM signal and to convert analog signal to digital measure; these components are interfaced with the PC through a digital input/output board by means of a serial data transmission. The microcontrollers are not in charge for the control of the movements, but are only used for interface between a software control system and the hardware. The controller has been implemented using RTAI [4], a real time application interface for Linux developed at Aerospace Engineering department of Politecnico di Milano. Section III contains a better description of the circuit and its interactions with the computer.

Two different control systems have been tested on the robot, a standard one, based on inverted kinematics and PID control, and the neural network based control for which the hexapod has been developed. The standard control system makes the legs follow a pre-stated motion pattern, and it can be improved with the use of *reflexes*, a technique which lets the control to better adapt to the external environment. The neural control system is based on the non-linearity of the CTRNN's, which can generate functional motion patterns as stable limit cycles. Section IV gives more details on how these control systems work, while section V will show some of the results obtained, with special attention to the fault tolerance properties of the neural control system. In section VI conclusions will be drawn comparing the effectiveness and the robustness of the two methods.

II. PROTOTYPE MECHANICAL DESIGN

The shape of the robot is meant to resemble the one of a six-legged insect, like an ant or a beetle and, even if almost all the biological insects have optimized shapes (e.g. often the backward legs are stronger as they are specialized in thrusting the animal) it has been decided that all the legs of the robot would have the same layout to achieve a simpler design. The legs are attached to a central body of parallelepipedal shape which contains the electronic circuits that drive the motors, forming two symmetrical rows of legs with respect to the longitudinal axis, resembling the biological features which shows a preferred direction for walking. In Fig. 1 the overall prototype is shown.

A leg is made up of two segments, and it is operated only by rotational joints (no sliding parts) as in insect limbs. The first segment of the leg is attached to the body forming a

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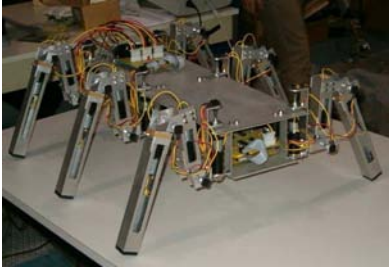


Fig. 1. Prototype

hip, and the second is attached to the first making a *knee*. The *hip* has two degrees of freedom which consist in a rotation around a vertical axis (γ , which allows the forward and backward motion of the leg) and around an horizontal longitudinal axis (β , which allows the leg to be raised and lowered). The *knee* has one rotational degree of freedom (α) which allows the leg to contract or to extend. In Fig. 2 it is possible to see the layout of these degrees of freedom.



Fig. 2. Leg configuration and angle definition.

In total, each leg owns 3 rotational degrees of freedom and will require 3 motors for actuation. The grand total is 18 degrees of freedom and 18 motors for the whole robot.

The leg segments are made of aluminium structural shapes in order to achieve both stiffness and lightweight. The empty space inside the legs is used for hosting 2 of the 3 leg motors, thus reducing the overall encumbrance of the limb. The outcome of this choice is that it compels the use of conical gears for actuating the *knee* joint (α) and the horizontal axis *hip* rotation (β); in fact, the torque output of the motors is parallel to the leg segments and it has to be turned into a perpendicular torque in order to be useful for actuation. The last motor, the one which moves the vertical axis *hip* degree of freedom (γ), is hosted in the body; the joint has been realized just constraining the motor shaft to the leg, while the motor case is fixed to the body.

The motors have been chosen on the base of static equilibrium considerations, i.e. assuring that they would be able to supply the minimal torque needed for sustainance with 3 legs touching the ground. The highest torque requirement is for β joint; however the motor used is the same for all the degrees of freedom in order to reduce the complexity of the design.

The table I shows some of the overall features of the prototype.

TABLE I
FEATURE OF THE PROTOTYPE

Weight	7.5 kg
Length	60 cm
Width (legs retracted)	45 cm
Maximum Continuous Motor Torque	2 Nm
Motor Feed	12 V
Average Power Consumption	19 W
Maximum Power Consumption	30 W

III. PROTOTYPE ELECTRONIC DESIGN

As the prototype has been designed for use in a laboratory it has been decided to leave the power generation and the computer responsible of control actions computation off-board. In this way the robot is not autonomous, however the mechanical design has been performed considering a weight of the robot which will allow to add a battery and a small on-board computer to allow autonomous operation.

The robot has a simple electronic circuit which allows to drive the motors under the command of the external PC. In order to realise a feedback control, the measure of the leg angles must be acquired, and this has required the installation of rotation sensors on each of the 18 joints. The circuit is also responsible for the interface between the sensors and the control software on the PC. Some integrated circuits are used for performing the functions of driving the motors, acquiring the sensors measure and exchanging all the necessary information in both ways with a PC.

The electronic design is modular, as the 6 legs are controlled by 6 independent identical circuits which both make the design simpler and ensure that a failure which would make a leg stop working would not affect the others.

A. Actuators Drive

The motor drive should allow to control the position of each joint and is achieved with a torque drive, which can be obtained regulating the current flowing through the motors.

The simplest way to do this kind of control is linking the motor to a *full bridge circuit* under the command of a PWM (Pulse Width Modulation) signal. The full bridge allows the motor to operate in both ways, at a constant feed voltage (which can be inverted to get motor reversal), in an *on-off* fashion; two logic inputs of the full bridge determine if the bridge is open (0 and 0, i.e. no current flowing), or closed in one way (0 and 1) or the other (1 and 0). The current (and so the torque) can be regulated modulating at high frequency the signal which opens and closes the full bridge, that is, using a PWM signal. A PWM signal is a periodic signal defined by its duty cycle percentage parameter, that indicates for which fraction of the period the signal is a logic 1, while it is a 0 for the rest of the time.

The full bridge is made up with the use of a commercial IC, the L6203 driver (one per motor) while the PWM signal can be efficiently generated by a microcontroller. It has been decided to use six PIC18F4431 microcontrollers, one per leg, which generate 3 PWM signals each. The microcontrollers

are able to communicate with the PC, they will receive from it the values of duty cycle which are to be executed.

There are two kinds of possible PWM drives, and both have been implemented:

- **Magnitude and sign PWM:** the driver inputs are a PWM signal and a constant 0 signal; the duty cycle determines the power of the action, while the direction of the torque depends on which of the two signals is 0 (e.g.: d.c.=0%: motor still; d.c.=100%: motor at full strength).

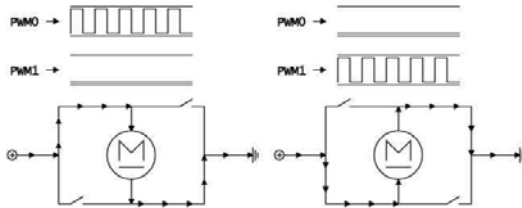


Fig. 3. Magnitude and sign PWM effect on full drive circuit

- **Locked anti-phase PWM:** the driver inputs are a PWM signal and its logic negation; this means that the full bridge is always closed in one direction or the other (e.g.: d.c.=50%: motor still; d.c.=100%: motor at full in one direction; d.c.=0%: motor at full strength in the other direction).

The best choice is the use of magnitude and sign PWM as the locked anti-phase PWM, even for low levels of torque required, causes a significant power consumption due to the oscillating currents flowing in opposite directions through the motor. This effect is strongly undesirable in a robot for space exploration and in the foresight of an upgrade of the prototype which will have to rely on battery for autonomy.

B. Sensors Acquisition

The control system of the robot needs the measures of the angles in order to perform a closed-loop control on the angles. The sensors chosen for this task are 18 potentiometers, one per joint, mounted coaxially on the joints thanks to a small aluminium support.

The potentiometers output are analog voltage signals which are acquired by the PIC18F4431 microcontrollers that provide the analog to digital conversion. Each microcontroller gets the measures of all the angles of one leg. It is convenient to use the microcontrollers for analog acquisition because it allows to use the same communication channels between the microcontroller's and the PC for transferring the data to the processing unit; if an analog input board were used, it would be necessary to use other 18 channels going from the robot to the PC, increasing the complexity of the system.

C. Computer Interface

The microcontrollers have the purpose of generating the PWM signals for motor drive and acquiring the angle measures, but the control system is not installed on them. A PC can grant better computing abilities than the microcontrollers, and it can also offer to the user a friendly control

interface; as the purpose of the prototype is the test of many different control systems, the use of a control system implemented on PC could allow a rapid switching between different type of controllers which are going to be tested.

The robot control system is hence a software executed on a computer; the computer communicates with the microcontrollers receiving from them the values of the angles and commanding the duty cycles to be executed by each motor. The whole computation for the control is thus executed by the PC only.

The communication between the microcontroller's and the PC is done through a digital input/output board; each microcontroller is connected to the board with two digital channels, one for input and one for output, leading to a total of 12 communication channels between the computer and the robot. The communication follows a serial protocol which has been developed ad hoc for this problem and in which the computer plays a *master* role and it communicates at the same time with all the microcontroller's, with the stream of data flowing in both ways altogether, thus reducing the communication time. The total bitrate of the communication is about 60 kbps, and it would allow the control to be executed at a frequency of up to 700 Hz (which is much higher than necessary).

The control system is a process coded in C++ running on a PC on which a Linux operative system is installed and made Real Time through the use of RTAI (Real Time Application Interface), which has been developed at the Department of Aerospace Engineering of Politecnico di Milano. The control system requires real time abilities for keeping the serial communication synchronicity and for performing the control at a constant frequency and is executed at a frequency of 25 Hz, as the system dynamics is very slow and intrinsically stable. This frequency is much slower than the maximum allowed by the PC, so it does not overload the machine.

IV. CONTROL SYSTEM

On this Prototype two different control system schemes have been tested:

- a classic PID (Proportional Integral Derivative) control system, which allows to follow the angle positions necessary for walking that have been generated with inverted kinematics algorithms.
- a control system based on an Artificial Neural Network in which the motor commands are generated by a CTRNN (Continuous Time Recurrent Neural Network) with the angle measures as input feedback; the weights and parameters of the network are evolved with a Genetic Algorithm.

Each system has its advantages and drawbacks, which will be discussed in section VI.

A. Gait Models

The robot walks using a coordinated leg motion pattern that can be observed in insects. Each pattern is defined by the period of each step, the fraction of the period in which the leg touches the ground and the relative phase between

the motion of the legs. There are two main gait patterns that can be observed in nature:

- The Tripod gait pattern which is the faster one (see Fig. 4). At least 3 legs are always touching the ground (either 2, 3 and 6 or 1, 4 and 5) granting the robot stability. Legs 2, 3 and 6 are half period behind 1, 4 and 5.
- The Wave gait pattern which is slower but more stable (see Fig. 5). The minimum number of legs touching the ground is always more than 3 and it depends on the ratio between the ground run and the period (in the case shown in figure there are always at least 4 legs down). Each leg has a phase which differs of $\frac{1}{6}$ of period from another, in the order: 1, 4, 5, 2, 3, 6.

These gait patterns are useful for maintaining the leg coordination and rhythm and can be used to obtain all the movements desired, from rectilinear forward/backward walk, to side walks and rotations on site.

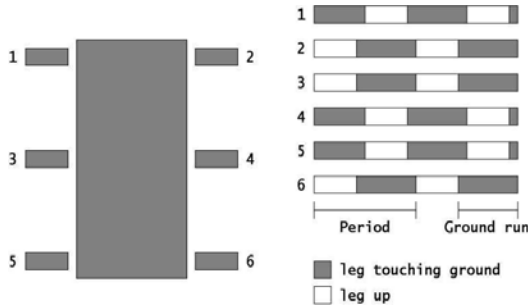


Fig. 4. Tripod gait

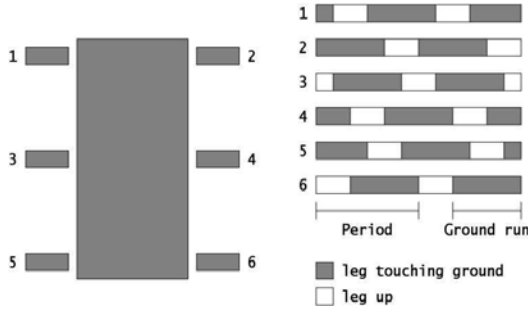


Fig. 5. Wave gait

B. PID Feedback Control

The system is based on a PID control that operates in the joint-space. The reference is generated during time with inverted kinematics, in order to achieve the gait pattern and motion type desired. The control system acts a feedback action (in terms of motor duty cycle) on the difference between the desired angles and the actual angles, and on its derivative and integral. The PID gains have been chosen on the basis of considerations on a simplified second order model of the system (i.e. the motion of a massive body with damping).

The baseline of this kind of control only makes the legs follow a pre-stated pattern, and it is poorly able to react

to external stimuli or unexpected occurrences. For example, the control would not be influenced by the fact that a leg that should be on ground is not touching it because of the irregular ground conformation (e.g. a hole), as long as the leg end is in the position stated.

The control system can be improved providing that the chased trajectory can change if special events occur; these behaviors can be called *reflexes*. A simple reflex has been implemented in the case of a leg which is expected to touch the ground and is not doing it, as the control systems lowers its position until ground is reached. The robot has no contact sensors on the leg end, so the fact that the leg is touching or not the ground is derived from the integral part of the control, which indicates whether the leg is contrasting a constant force (i.e. the force necessary for sustaining the robot) or not (i.e., the leg is just leaning out).

Fig. 6 shows the action of the reflex implemented when a leg encounters a hole.

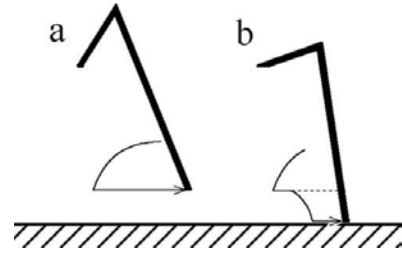


Fig. 6. Leg "a" is controlled with plain feedback on a reference trajectory, and its motion is not affected by the environment. Leg "b" has a reflex control, and if it "feels" that it is not touching the ground, it descends until contact.

C. CTRNN Application

A CTRNN is an artificial neural network with an internal dynamics; it is a set of integrators (states or *neurons*) linked together with differential equations. The network has inputs and outputs, and the outputs depend on the whole history of the inputs. The dynamic of the network is non-linear.

In formulas, a CTRNN has m inputs, n neurons (states) and l outputs. Defining the following matrices:

$[X]_{n \times 1}$	state vector	$[I]_{m \times 1}$	input vector
$[Y]_{l \times 1}$	output vector	$[W]_{n \times x}$	weight matrix
$[P]_{m \times m}$	input matrix	$[K]_{l \times n}$	output matrix
$[\tau]_{n \times 1}$	time constants vector	$[\theta]_{n \times 1}$	bias vector

the dynamics of the neuron states is expressed by the equations:

$$\dot{x}_i \tau_i = -x_i + \sum_{j=1}^n W_{ij} \sigma_j(x_j) + \sum_j P_{ij} I_j \quad (1)$$

$$\sigma_j(z) = \frac{1}{(1 + e^{(-z - \theta_j)})} \quad (2)$$

while the outputs are:

$$y_i = \sum_{j=1}^n K_{ij} x_j \quad (3)$$

A network of this kind can generate control outputs with complex behavior (i.e. oscillating signals), and they can be used as a control system for a legged robot, in analogy with a nervous system controlling the gait of an insect. This means that the network inputs are the measures of the angles, and the outputs are the motor duty cycles.

There is no reference signal this time as the neural control system is linked to the robot mechanics and can generate a complex dynamic based on the fact that the internal states of the neurons can converge on limit cycles. Realising a neural control system means finding the correct parameters which generate a significative cyclical motion as a stable limit cycle.

The neural network has been trained with a Genetic Algorithm [5] in which the fitness function is computed through a numerical simulation of the network over a fixed amount of time which allows to evaluate the behaviour of the network. The aim of the training is to find the parameters of the network which allow to generate one of the gaits for a rectilinear motion. In order to do that the fitness function is structured in a way that allows the solution to reach the final desired behaviour step by step, searching for the features of the solution one at a time. In particular the first feature searched is the periodicity of the motion, and only once that the solution found by the GA are periodic the fitness function is switched to a second feature, which can be the amount of travelled space or the mean velocity of the motion.

A network of 30 neurons has been successfully trained to generate a tripod walk, and it has been effective also in the real environment. The result obtained in the simulation can be translated into the real robot if care is taken that the network dynamic is not overfitted to the simulated environment, that is, if the inner rhythmical dynamic of the network is in part independent from the environment.

In order to ease the work of the genetic algorithm, the problem of finding the weights of the network has not been solved in a single shot, but it has been divided into 4 smaller problems. A network of 3 neurons has been evolved for controlling the γ joint of a leg, then two other neurons and their links have been added for the control of β and α , thus obtaining a 5-neuron network controlling a single leg. The last step was to join six legs of this kind and evolve their links in order to obtain the coordinated motion of all the legs which reproduce a gait pattern.

In Fig. 7 The local network for the control of a single leg is shown. The circles represent the neurons and the thin lines indicate the connections (those due to W matrix) while the wide arrows show where the angle measures enter in the network.

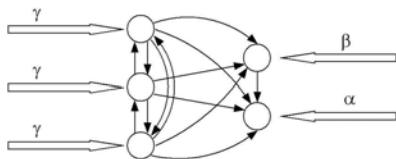


Fig. 7. The local network for the control of a single leg

V. RESULTS

A. Motion under a standard Control System

The PID control with inverted kinematics is highly precise and lets the robot do any kind of movement and walk in any direction, with full trajectory control. An high level control system (or a user, with the help of a joystick-like interface) can navigate the robot in any fashion needed at any speed, up to a maximum of about 6 cm/s.

In Fig. 8 the trajectory of a leg tip with respect to the attachment point of the leg during a fast tripod walk is shown; the thick dotted line shows the actual position, while the thin line is the reference position.

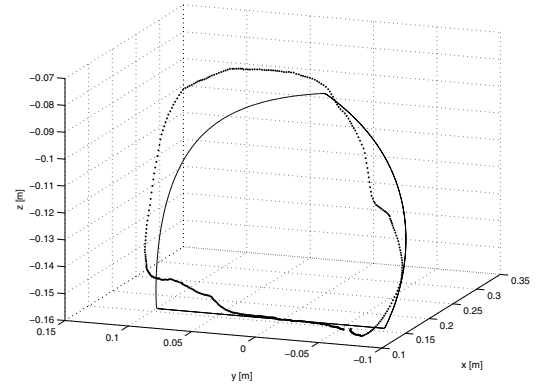


Fig. 8. Movement of the leg end under the effect of PID control

B. Motion under the control of a Neural Network

The motion under the effect of a neural network is not as precise as the one generated with the previous control system. The leg movement is mainly composed by sine-like oscillations of the angles, well coordinated and able to keep the robot stable while walking. For this reason, the trajectory of the legs on ground is not exactly rectilinear, and the robot swings a little when the weights are redistributed in the gait (i.e., when a group of 3 legs is raised in the tripod gait.) In Fig. 9 the tip movement in the case of neural control is shown.

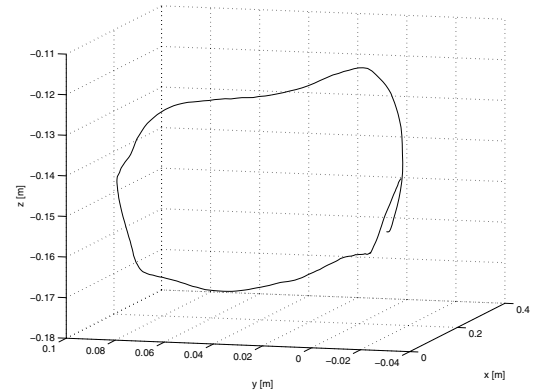


Fig. 9. The rhythmic step movement generated by the CTRNN.

An interesting property of the neural control is that the whole movement of the robot is a stable cycle. If one leg is

stopped for any reason, when it is released it returns at once to its former movement, restoring the coordination with the other legs. In Fig. 10 the reaction of the neural controller to a sudden stop of the γ joint can be seen. The leg movement suffers a deformation but returns to the usual cycle after being released. In Fig. 11 the evolution of the state of the neurons involved in the motion of γ joint is reported for the same situation. The states of the neurons start from the arbitrary value of 0 and reach a stable limit cycle after a short transient (to the bottom). When the leg is halted, the cycle is warped and shrunk (thick dotted line), but it returns as before after release.

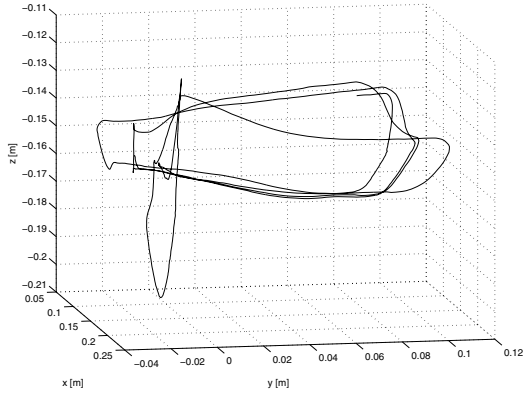


Fig. 10. Motion of a leg under the control of a CTRNN

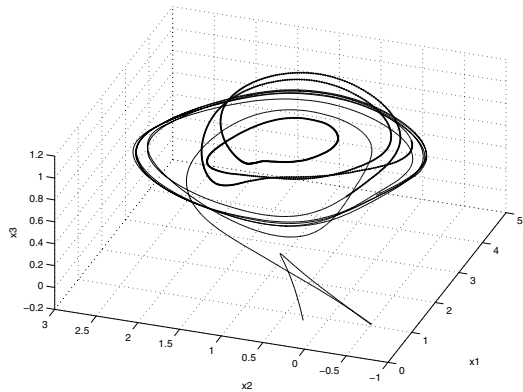


Fig. 11. The evolution of the state of the neurons involved in the motion of gamma joint

Another impressive property of the neural control is that it is able to endure a sensor failure without compromising the functionality of a leg. If one γ sensor is removed, after a transient, the leg manages to follow a new rythmical step motion, although warped. In Fig. 12 the tip position is shown, both in the case of a fully functional γ sensor (the dotted line) and in the case without the sensor on γ joint.

VI. CONCLUSIONS

The results of this work are interesting under many points of view. First of all, it has been shown that it is possible to run a complex device such as a hexapod robot under

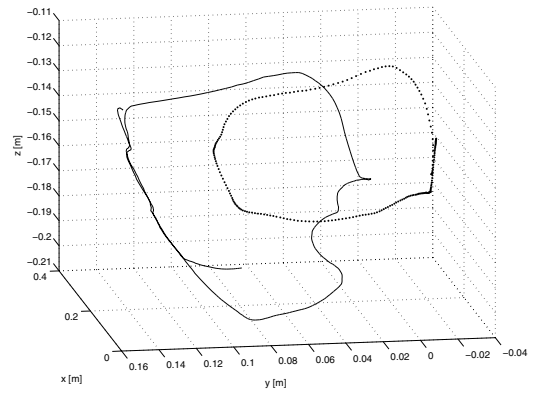


Fig. 12. Motion pattern generated by the neural network with sensor failure (solid line) and in nominal state (dotted line)

the control of an artificial neural network evolved through simulations. Secondly, it has allowed to characterize the differences between a classic control system and a CTRNN-based system.

The classic control system ensures reliability, total control and precision. The technique of just making the legs follow a pre-stated trajectory is not able to adapt the movements to the external environment, but with the use of reflexes the control can be refined as far as desired. The neural control is smoother and better able to react to external stimuli without any programmed reflexes as it will be able to adapt (up to a certain limit) the leg trajectory to the environment. Moreover, it has been proved to be greatly fault tolerant, as it can continue to execute an acceptable motion even with a sensor failure. As a drawback, the neural control cannot achieve a control as precise as the one ensured by inverted kinematics; a small CTRNN like the ones considered in this work (5 neurons per leg) is more suitable to generate an approximate rithmical motion. Further research could discover if the use of a more complex network might approximate any movement with the desired tolerance.

Finally it can be said that the results obtained are encouraging as the relfexes needed to face a rough terrain have been obtained both with a classical control and with the neural network. More work is needed to train a neural network to face a general rough terrain, while the classical control is able to succesfully face obstacles which are lower of the height of the bottom of the robot.

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